

Nash Equilibrium

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March 21, 2012

Rationalizability as a Solution Concept

- Rationalizability makes very weak predictions:

		Player 2	
		B	F
Player 1	F	0, 0	2, 1
	B	1, 2	0, 0

Figure: Battle of Sexes

- Each player plays his/her best response to a subjective belief to the other player's strategy.
- Rationalizability does not impose any restrictions on these beliefs.
- In contrast, *Nash Equilibrium* requires that the player's belief are correct in equilibrium.

Definition: Nash Equilibrium

A mixed strategy profile σ^* is a Nash equilibrium if, for all players i

$$u_i(\sigma_i^*, \sigma_{-i}^*) \geq u_i(s_i, \sigma_{-i}^*) \quad \forall s_i \in S_i$$

NE is strict if the above inequality is strict.

- “Consistent prediction” of how the game will be played.
 - given that beliefs are correct, nobody has incentives to deviate unilaterally.

Example

Prisoner's Dilemma

		Player 2	
		C	NC
Player 1	C	$-9, -9$	$0, -10$
	NC	$-10, 0$	$-1, -1$

Figure: Prisoner's Dilemma

Nash Equilibrium v.s. ISD and Rationalizability

- Any Nash equilibrium strategy survives ISD.
 - $R_i \subseteq \sum_i^\infty$.
- A unique prediction from ISD is also a unique NE.
- The reverse, whoever is not true.

Example

		Player 2	
		B	F
Player 1	F	0, 0	2, 1
	B	1, 2	0, 0

Figure: Battle of Sexes

Best Response Correspondence

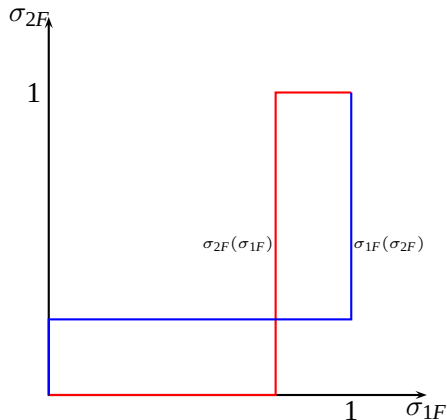
- The best response $\sigma_{1F}(\sigma_{2F})$ of player 1 is

$$\sigma_{1F}(\sigma_{2F}) = \begin{cases} 0 & \text{if } \sigma_{2F} < \frac{1}{3} \\ [0, 1] & \text{if } \sigma_{2F} = \frac{1}{3} \\ 1 & \text{if } \sigma_{2F} > \frac{1}{3} \end{cases}$$

- The best response $\sigma_{2F}(\sigma_{1F})$ of player is

$$\sigma_{2F}(\sigma_{1F}) = \begin{cases} 0 & \text{if } \sigma_{1F} < \frac{2}{3} \\ [0, 1] & \text{if } \sigma_{1F} = \frac{2}{3} \\ 1 & \text{if } \sigma_{1F} > \frac{2}{3} \end{cases}$$

Best Response Correspondence



Existence of Nash Equilibrium

- A pure strategy equilibrium does not always exist.
 - zero-sum games

		Player 2	
		H	T
Player 1	H	1, -1	-1, 1
	T	-1, 1	1, -1

Figure: Matching Pennies

Existence of NE in Mixed Strategies

Theorem

Every finite strategic-form game has a mixed-strategy Nash Equilibrium.

Proof

Let $r_i(\sigma_{-i})$ denote i 's best response correspondence:

$$r_i(\sigma_{-i}) \equiv \{\sigma_i \in \Sigma_i \mid \forall s'_i \in S_i, u_i(\sigma_i, \sigma_{-i}) \geq u_i(s'_i, \sigma_{-i})\}$$

The aggregate best response correspondence

$$r(\sigma) \equiv r_1(\sigma_{-1}) \times \dots \times r_I(\sigma_{-I}) \subseteq \Sigma$$

Then a strategy profile σ^* is a NE if $\sigma^* \in r(\sigma)$ (fixed point).

Existence of a Fixed Point

Kakutani's Fixed Point Theorem

Sufficient conditions for existence of a fixed point (Kakutani):

- 1) Σ is a compact, convex and nonempty.
 - 2) $r(\sigma)$ - nonempty and convex.
 - 3) $r(\sigma)$ is upper hemi-continuous.
- Compactness of Σ : closed and bounded.
 - Convexity of Σ : contains all possible mixing over player's pure strategies.
 - Non-emptiness of $r(\sigma)$: player's payoff function u_i is continuous in his own mixed strategy. Continuous function on compact sets always attains a maxima.
 - Convexity of $r(\sigma)$: if $\sigma_i, \sigma'_i \in r(\sigma_i)$, then $\lambda\sigma_i + \lambda\sigma''_i \in r(\sigma_i)$. Why?

Existence of a Fixed Point

- Upper hemi-continuity of $r(\sigma)$: if
 - a sequence $(\sigma_i^k, \sigma_{-i}^k) \rightarrow (\sigma_i, \sigma_{-i})$ and
 - $\sigma_i^k \in r(\sigma_{-i}) \forall k$
then $\sigma_i \in r(\sigma_{-i})$.

Since u_i is continuous and $(\sigma_i^k, \sigma_{-i}^k) \rightarrow (\sigma_i, \sigma_{-i})$, this must hold.

- The existence of NE extends to infinite game with continuous payoffs as well.

References

- Fudenberg and Tirole- Chapter 1,2